

T3 UV one-loop RGE term comparison

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2 - 1)/4$
$T(d)$	$d(d^2 - 1)/12$
$C_{LLS_1 S_2}^{ij:AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \rightarrow g$ conversion as in the previous report.

$\beta_{g_1}^{(1)}$

Quantity: g_1

$$16\pi^2\,\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$$

Column Representative term structure	
T1	$\frac{163g_1^3}{6}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{163g_1^3}{6}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{55g_1^3}{6}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{59g_1^3}{6}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\frac{25g_1^3}{2}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\frac{43g_1^3}{6}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\frac{25g_1^3}{2}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{41g_1^3}{2}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{43g_1^3}{6}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{41g_1^3}{2}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{59g_1^3}{6}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{55g_1^3}{6}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{163g_1^3}{6}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{179g_1^3}{6}$

Model	α	S_1	S_2	F	T1
T3-E-m2	-2	$(3, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	$\frac{59g_1^3}{6}$
T3-E-p0	0	$(3, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	$\frac{59g_1^3}{6}$
T3-E-p2	2	$(3, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	$\frac{179g_1^3}{6}$

$\beta_{g_2}^{(1)}$
Quantity: g_2

$$16\pi^2\,\mu\,\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column		Representative term structure
T1	$-\frac{g_2^3}{2}$	
T2	$-\frac{17g_2^3}{6}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{g_2^3}{2}$	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{g_2^3}{2}$	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{g_2^3}{2}$	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	$-\frac{17g_2^3}{6}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	—	$-\frac{17g_2^3}{6}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	$-\frac{17g_2^3}{6}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	$\frac{31g_2^3}{6}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	—	$\frac{7g_2^3}{6}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	$\frac{31g_2^3}{6}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-\frac{g_2^3}{2}$	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$-\frac{g_2^3}{2}$	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-\frac{g_2^3}{2}$	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$\frac{g_2^3}{6}$	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$\frac{g_2^3}{6}$	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$\frac{g_2^3}{6}$	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$\frac{g_2^3}{6}$	—

$\beta_{g_3}^{(1)}$
Quantity: g_3

$$16\pi^2\,\mu\,\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column		Representative term structure	
T1		$-7g_3^3$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-7g_3^3$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-7g_3^3$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-7g_3^3$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$-7g_3^3$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-7g_3^3$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$-7g_3^3$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$-7g_3^3$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$-7g_3^3$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$-7g_3^3$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-7g_3^3$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$-7g_3^3$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-7g_3^3$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-7g_3^3$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-7g_3^3$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-7g_3^3$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-7g_3^3$

$$16\pi^2\,\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$$

Terms 1 of 3

Column		Representative term	structure
T1	3	$\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	
T2	+	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	
T3	+3	$\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	3 Tr $\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_u)_{ij}$	+ Tr $\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_u)_{ij}$	+3 Tr $\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column		Representative term structure
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	
T3	$-8g_3^2\ (Y_u)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column		Representative term structure
T1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-A-m2	-2	$(1, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-A-p0	0	$(1, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(1, -1)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(1, 1)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(3, -1)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(3, 0)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(3, 1)$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D-m2	-2	$(3, -1)$	$(1, 0)$	$(2, -\frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D-p0	0	$(3, 0)$	$(1, 1)$	$(2, \frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D-p2	2	$(3, 1)$	$(1, 2)$	$(2, \frac{3}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E-m4	-4	$(3, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E-m2	-2	$(3, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E-p0	0	$(3, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E-p2	2	$(3, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$
Quantity: Y_d

$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$

Terms 1 of 3

Column		Representative term structure
T1		$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	
T3	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column		Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	
T3	$-8g_3^2\ (Y_d)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column		Representative term structure
T1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

$16\pi^2\,\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 3

Column		Representative term structure
T1	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	
T3	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2,-\frac{3}{2})$	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2,-\frac{1}{2})$	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2,\frac{1}{2})$	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1, -1)	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1, 0)	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1, 1)	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3, -1)	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3, 0)	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3, 1)	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-D-m2	-2	(3, -1)	(1, 0)	$(2,-\frac{1}{2})$	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-D-p0	0	(3, 0)	(1, 1)	$(2,\frac{1}{2})$	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-D-p2	2	(3, 1)	(1, 2)	$(2,\frac{3}{2})$	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2,-\frac{3}{2})$	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2,-\frac{1}{2})$	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2,\frac{1}{2})$	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2,\frac{3}{2})$	—	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$

Quantity: Y_e

Terms 3 of 3

Column		Representative term structure	
T1		$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$

$\beta_{y_1}^{(1)}$

Quantity: y_1

$$16\pi^2\,\mu\,\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$$

Terms 1 of 5

Column	Representative term structure	
T1	$2T_{\nu}^{(1)}\left(y_1\right)_{ij}$	
T2	$+\frac{3}{4}\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}$	
T3	$+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$2T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{3}{4}\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$2T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{3}{4}\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$2T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{3}{4}\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: y_1

Terms 2 of 5

Column		Representative term structure
T1		$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$
T2		$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$
T3		$-\frac{15}{2} g_1^2 \left(y_1 \right)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$\left(1, -2 \right)$	$\left(3, -1 \right)$	$\left(2, -\frac{3}{2} \right)$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-A-m2	-2	$\left(1, -1 \right)$	$\left(3, 0 \right)$	$\left(2, -\frac{1}{2} \right)$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-A-p0	0	$\left(1, 0 \right)$	$\left(3, 1 \right)$	$\left(2, \frac{1}{2} \right)$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-B-m3	-3	$\left(2, -\frac{3}{2} \right)$	$\left(2, -\frac{1}{2} \right)$	$\left(1, -1 \right)$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-B-m1	-1	$\left(2, -\frac{1}{2} \right)$	$\left(2, \frac{1}{2} \right)$	$\left(1, 0 \right)$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-B-p1	1	$\left(2, \frac{1}{2} \right)$	$\left(2, \frac{3}{2} \right)$	$\left(1, 1 \right)$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-C-m3	-3	$\left(2, -\frac{3}{2} \right)$	$\left(2, -\frac{1}{2} \right)$	$\left(3, -1 \right)$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-C-m1	-1	$\left(2, -\frac{1}{2} \right)$	$\left(2, \frac{1}{2} \right)$	$\left(3, 0 \right)$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-C-p1	1	$\left(2, \frac{1}{2} \right)$	$\left(2, \frac{3}{2} \right)$	$\left(3, 1 \right)$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-D-m2	-2	$\left(3, -1 \right)$	$\left(1, 0 \right)$	$\left(2, -\frac{1}{2} \right)$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-D-p0	0	$\left(3, 0 \right)$	$\left(1, 1 \right)$	$\left(2, \frac{1}{2} \right)$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-D-p2	2	$\left(3, 1 \right)$	$\left(1, 2 \right)$	$\left(2, \frac{3}{2} \right)$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-E-m4	-4	$\left(3, -2 \right)$	$\left(3, -1 \right)$	$\left(2, -\frac{3}{2} \right)$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-E-m2	-2	$\left(3, -1 \right)$	$\left(3, 0 \right)$	$\left(2, -\frac{1}{2} \right)$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-E-p0	0	$\left(3, 0 \right)$	$\left(3, 1 \right)$	$\left(2, \frac{1}{2} \right)$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-E-p2	2	$\left(3, 1 \right)$	$\left(3, 2 \right)$	$\left(2, \frac{3}{2} \right)$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_1^2 \left(y_1 \right)_{ij}$

Quantity: y_1

Terms 3 of 5

Column		Representative term structure
T1		$T_{\nu}^{(1)}(y_1)_{ij}$
T2		$+\frac{1}{2}((y_2(y_2)^{\dagger}y_1)_{ij}+(Y_e(Y_e)^{\dagger}y_1)_{ij}+3(y_1(y_1)^{\dagger}y_1)_{ij})$
T3		$\text{Tr}(y_1(y_2)^{\dagger})(y_2)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^{\dagger}y_1)_{ij}+(Y_e(Y_e)^{\dagger}y_1)_{ij}+3(y_1(y_1)^{\dagger}y_1)_{ij})$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+T_{\nu}^{(1)}(y_1)_{ij}$	—	$\text{Tr}(y_1(y_2)^{\dagger})(y_2)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^{\dagger}y_1)_{ij}+(Y_e(Y_e)^{\dagger}y_1)_{ij}+3(y_1(y_1)^{\dagger}y_1)_{ij})$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^{\dagger}y_1)_{ij}+(Y_e(Y_e)^{\dagger}y_1)_{ij}+3(y_1(y_1)^{\dagger}y_1)_{ij})$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^{\dagger}y_1)_{ij}+(Y_e(Y_e)^{\dagger}y_1)_{ij}+3(y_1(y_1)^{\dagger}y_1)_{ij})$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^{\dagger}y_1)_{ij}+(Y_e(Y_e)^{\dagger}y_1)_{ij}+3(y_1(y_1)^{\dagger}y_1)_{ij})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$T_{\nu}^{(1)}(y_1)_{ij}$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$T_{\nu}^{(1)}(y_1)_{ij}$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$T_{\nu}^{(1)}(y_1)_{ij}$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$T_{\nu}^{(1)}(y_1)_{ij}$	—	—

Quantity: y_1

Terms 4 of 5

Column	Representative term structure
T1	$+\frac{1}{2}(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T2	$\frac{3}{2}T_{\nu}^{(1)}\left(y_1\right)_{ij}$
T3	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{1}{2}(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\frac{3}{2}T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij})$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+\frac{3}{2}T_{\nu}^{(1)}\left(y_1\right)_{ij}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\frac{3}{2}T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij})$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij})$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij})$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij})$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij})$

Quantity: y_1

Terms 5 of 5

Column	Representative term structure
T1	$\frac{3}{2} \text{Tr}\left(y_1 \left(y_2\right)^\dagger\right) \left(y_2\right)_{ij}$
T2	$+\frac{1}{4} (2 \left(y_1 \left(y_2\right)^\dagger y_2\right)_{ij} + 3 \left(y_2 \left(y_2\right)^\dagger y_1\right)_{ij} + 2 \left(Y_e \left(Y_e\right)^\dagger y_1\right)_{ij} + 5 \left(y_1 \left(y_1\right)^\dagger y_1\right)_{ij})$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\frac{3}{2} \text{Tr}\left(y_1 \left(y_2\right)^\dagger\right) \left(y_2\right)_{ij}$	$+\frac{1}{4} (2 \left(y_1 \left(y_2\right)^\dagger y_2\right)_{ij} + 3 \left(y_2 \left(y_2\right)^\dagger y_1\right)_{ij} + 2 \left(Y_e \left(Y_e\right)^\dagger y_1\right)_{ij} + 5 \left(y_1 \left(y_1\right)^\dagger y_1\right)_{ij})$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{y_2}^{(1)}$

Quantity: y_2

$$16\pi^2\,\mu\,\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$T_{\nu}^{(2)}(y_2)_{ij}$
T2	$+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\,(y_2\ (y_2)^{\dagger}\ y_2)_{ij})$
T3	$-\frac{9}{2}g_2^2(y_2)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$T_{\nu}^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\,(y_2\ (y_2)^{\dagger}\ y_2)_{ij})$	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$T_{\nu}^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\,(y_2\ (y_2)^{\dagger}\ y_2)_{ij})$	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$T_{\nu}^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\,(y_2\ (y_2)^{\dagger}\ y_2)_{ij})$	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$T_{\nu}^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\,(y_2\ (y_2)^{\dagger}\ y_2)_{ij})$	$-\frac{9}{4}g_2^2(y_2)_{ij}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+T_{\nu}^{(2)}(y_2)_{ij}$	—	$-\frac{9}{4}g_2^2(y_2)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$T_{\nu}^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\,(y_2\ (y_2)^{\dagger}\ y_2)_{ij})$	$-\frac{9}{4}g_2^2(y_2)_{ij}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	$-\frac{33}{4}g_2^2(y_2)_{ij}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	$-\frac{33}{4}g_2^2(y_2)_{ij}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	$-\frac{33}{4}g_2^2(y_2)_{ij}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$T_{\nu}^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$T_{\nu}^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$T_{\nu}^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$T_{\nu}^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$

Quantity: y_2

Terms 2 of 5

Column		Representative term structure
T1	$-\frac{15}{2}g_1^2(y_2)_{ij}$	
T2	$\text{Tr}\left(y_2(y_1)^\dagger\right)(y_1)_{ij}$	
T3	$+\frac{1}{2}((y_1(y_1)^\dagger y_2)_{ij}+2(y_2(y_1)^\dagger y_1)_{ij}+(Y_e(Y_e)^\dagger y_2)_{ij}+3(y_2(y_2)^\dagger y_2)_{ij})$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{15}{2}g_1^2(y_2)_{ij}$	—	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$-\frac{15}{4}g_1^2(y_2)_{ij}$	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-\frac{3}{4}g_1^2(y_2)_{ij}$	$\text{Tr}\left(y_2(y_1)^\dagger\right)(y_1)_{ij}$	$+\frac{1}{2}((y_1(y_1)^\dagger y_2)_{ij}+2(y_2(y_1)$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$-\frac{15}{4}g_1^2(y_2)_{ij}$	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$-\frac{15}{4}g_1^2(y_2)_{ij}$	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$-\frac{3}{4}g_1^2(y_2)_{ij}$	—	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$-\frac{15}{4}g_1^2(y_2)_{ij}$	—	—
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-\frac{15}{2}g_1^2(y_2)_{ij}$	—	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-\frac{15}{2}g_1^2(y_2)_{ij}$	—	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-\frac{15}{2}g_1^2(y_2)_{ij}$	—	—

Quantity: y_2

Terms 3 of 5

Column	Representative term structure
T1	$\frac{3}{2}T_{\nu}^{(2)}(y_2)_{ij}$
T2	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij})$
T3	$\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)(y_1)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$\frac{3}{2}T_{\nu}^{(2)}(y_2)_{ij}$	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij})$	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+\frac{3}{2}T_{\nu}^{(2)}(y_2)_{ij}$	—	$\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)(y_1)_{ij}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$\frac{3}{2}T_{\nu}^{(2)}(y_2)_{ij}$	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij})$	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+6\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij})$	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+6\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij})$	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+6\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij})$	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+6\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij})$	—

Quantity: y_2

Terms 4 of 5

Column		Representative term structure
T1	$\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij})$	
T2	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$	
T3	$+\frac{3}{4}\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij})$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$	$+\frac{3}{4}\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$	$+\frac{3}{4}\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$	$+\frac{3}{4}\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: y_2

Terms 5 of 5

Column		Representative term structure
T1	$+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}$	
T2	$+\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	—	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	—	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	—
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}$		$+\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}$		$+\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}$		$+\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	—	—

$\beta_{M_F}^{(1)}$

Quantity: M_F

$16\pi^2\mu\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$
T2	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$
T3	$-\frac{27}{2}g_1^2\left(M_F\right)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{27}{2}g_1^2\left(M_F\right)_{ij}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{3}{2}g_1^2\left(M_F\right)_{ij}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{3}{2}g_1^2\left(M_F\right)_{ij}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{3}{2}g_1^2\left(M_F\right)_{ij}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{3}{2}g_1^2\left(M_F\right)_{ij}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{27}{2}g_1^2\left(M_F\right)_{ij}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{27}{2}g_1^2\left(M_F\right)_{ij}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{3}{2}g_1^2\left(M_F\right)_{ij}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{3}{2}g_1^2\left(M_F\right)_{ij}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$	$-\frac{27}{2}g_1^2\left(M_F\right)_{ij}$

Quantity: M_F

Terms 2 of 5

Column	Representative term structure
T1	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$
T2	$+ \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$
T3	$-6g_1^2\ (M_F)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$	$+ \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$	$-6g_1^2\ (M_F)_{ij}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+ \left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$	$+ \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$	$-6g_1^2\ (M_F)_{ij}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	$-6g_1^2\ (M_F)_{ij}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	$-6g_1^2\ (M_F)_{ij}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: M_F

Terms 3 of 5

Column	Representative term structure
T1	$\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$
T2	$+ \left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$
T3	$+ \left((y_2)^T\ (y_2)^*\ M_F\right)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$	$+ \left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$	$+ \left((y_2)^T\ (y_2)^*\ M_F\right)_{ij}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: M_F

Terms 4 of 5

Column	Representative term structure
T1	$\frac{1}{2}((y_1)^T (y_1)^* M_F)_{ij} + (M_F (y_2)^{\dagger} y_2)_{ij}$
T2	$-12g_2^2 (M_F)_{ij}$
T3	$\frac{1}{2}((M_F)^T (y_1)^{\dagger} y_1)_{ij} + ((y_1)^T (y_1)^* M_F)_{ij} + ((M_F)^T (y_2)^{\dagger} y_2)_{ij} + ((y_2)^T (y_2)^* M_F)_{ij}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\frac{1}{2}(((y_1)^T (y_1)^* M_F)_{ij} + (M_F (y_2)^{\dagger} y_2)_{ij})$	$-12g_2^2 (M_F)_{ij}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	$\frac{1}{2}(((M_F)^T (y_1)^{\dagger} y_1)_{ij} + ((y_1)^T (y_1)^* M_F)_{ij})$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\frac{1}{2}(((y_1)^T (y_1)^* M_F)_{ij} + (M_F (y_2)^{\dagger} y_2)_{ij})$	$-12g_2^2 (M_F)_{ij}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\frac{3}{4}(((y_1)^T (y_1)^* M_F)_{ij} + (M_F (y_2)^{\dagger} y_2)_{ij})$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\frac{3}{4}(((y_1)^T (y_1)^* M_F)_{ij} + (M_F (y_2)^{\dagger} y_2)_{ij})$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\frac{3}{4}(((y_1)^T (y_1)^* M_F)_{ij} + (M_F (y_2)^{\dagger} y_2)_{ij})$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\frac{3}{4}(((y_1)^T (y_1)^* M_F)_{ij} + (M_F (y_2)^{\dagger} y_2)_{ij})$	—	—

Quantity: M_F

Terms 5 of 5

Column	Representative term structure
T1	$-6g_2^2((M_F)^T)_{ij} + (M_F)_{ij}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$-6g_2^2((M_F)^T)_{ij} + (M_F)_{ij}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—

$\beta_{m_1^2}^{(1)}$

Quantity: m_1^2

$16\pi^2\,\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$
T2	$-8\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$
T3	$-24g_1^2m_1^2$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	$-8\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-24g_1^2m_1^2$
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	$-8\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	$-8\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	$-6\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	$-6\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-24g_1^2m_1^2$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$

Quantity: m_1^2

Terms 2 of 5

Column		Representative term structure
T1		$+6\lambda_{12}^{(1)}m_2^2$
T2		$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$
T3		$-\frac{9g_2^2m_1^2}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+6\lambda_{12}^{(1)}m_2^2$	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+6\lambda_{12}^{(1)}m_2^2$	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	$+6\lambda_{12}^{(1)}m_2^2$	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	$+4\lambda_{12}^{(1)}m_2^2$	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$	$-\frac{9g_2^2m_1^2}{2}$
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$+4\lambda_{12}^{(1)}m_2^2$	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$	$-\frac{9g_2^2m_1^2}{2}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	$+4\lambda_{12}^{(1)}m_2^2$	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$	$-\frac{9g_2^2m_1^2}{2}$
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	$+4\lambda_{12}^{(1)}m_2^2$	—	$-\frac{9g_2^2m_1^2}{2}$
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$+4\lambda_{12}^{(1)}m_2^2$	—	$-\frac{9g_2^2m_1^2}{2}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	$+4\lambda_{12}^{(1)}m_2^2$	—	$-\frac{9g_2^2m_1^2}{2}$
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	$+2\lambda_{12}^{(1)}m_2^2$	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	$+2\lambda_{12}^{(1)}m_2^2$	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	$+2\lambda_{12}^{(1)}m_2^2$	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: m_1^2

Terms 3 of 5

Column	Representative term structure
T1	$-\frac{27g_1^2m_1^2}{2}$
T2	$-8\text{Tr}\Big(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\Big)$
T3	$-8\text{Tr}\Big(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\Big)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	$-\frac{27g_1^2m_1^2}{2}$	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$-\frac{3g_1^2m_1^2}{2}$	$-8\text{Tr}\Big(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\Big)$	$-8\text{Tr}\Big(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\Big)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	$-\frac{3g_1^2m_1^2}{2}$	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	$-\frac{27g_1^2m_1^2}{2}$	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$-\frac{3g_1^2m_1^2}{2}$	$-12\text{Tr}\Big(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\Big)$	$-12\text{Tr}\Big(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\Big)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	$-\frac{3g_1^2m_1^2}{2}$	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: m_1^2

Terms 4 of 5

Column	Representative term structure
T1	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$
T2	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$
T3	$-12g_2^2 m_1^2$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2 m_1^2$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2 m_1^2$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2 m_1^2$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2 m_1^2$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2 m_1^2$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2 m_1^2$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2 m_1^2$

Quantity: m_1^2

Terms 5 of 5

Column	Representative term structure
T1	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, $-\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, $-\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, $-\frac{3}{2}$)	(2, $-\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, $-\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, $-\frac{3}{2}$)	(2, $-\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, $-\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, $-\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, $-\frac{3}{2}$)	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, $-\frac{1}{2}$)	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$\beta_{m_2^2}^{(1)}$

Quantity: m_2^2

$16\pi^2\,\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$

Terms 1 of 5

Column		Representative term structure
T1	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	
T2	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	
T3	$-12g_2^2m_2^2$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^2m_2^2$
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^2m_2^2$
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^2m_2^2$
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	$-6\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	$-6\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	$-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	$-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	$-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^2m_2^2$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^2m_2^2$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^2m_2^2$
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^2m_2^2$

Quantity: m_2^2

Terms 2 of 5

Column	Representative term structure	
T1	$-6g_1^2m_2^2$	
T2	$+2\lambda_{12}^{(1)}m_1^2$	
T3	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-6g_1^2m_2^2$	$+2\lambda_{12}^{(1)}m_1^2$	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+2\lambda_{12}^{(1)}m_1^2$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-6g_1^2m_2^2$	$+2\lambda_{12}^{(1)}m_1^2$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$+4\lambda_{12}^{(1)}m_1^2$	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$+4\lambda_{12}^{(1)}m_1^2$	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$+4\lambda_{12}^{(1)}m_1^2$	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$+4\lambda_{12}^{(1)}m_1^2$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+4\lambda_{12}^{(1)}m_1^2$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+4\lambda_{12}^{(1)}m_1^2$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$+6\lambda_{12}^{(1)}m_1^2$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-6g_1^2m_2^2$	$+6\lambda_{12}^{(1)}m_1^2$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-24g_1^2m_2^2$	$+6\lambda_{12}^{(1)}m_1^2$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-6g_1^2m_2^2$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-6g_1^2m_2^2$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-24g_1^2m_2^2$	—	—

Quantity: m_2^2

Terms 3 of 5

Column	Representative term structure
T1	$-\frac{9g_2^2m_2^2}{2}$
T2	$-\frac{3g_1^2m_2^2}{2}$
T3	$-8\text{Tr}\Big(M_F\left(M_F\right)^*\left(y_2\right)^T\left(y_2\right)^*\Big)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$	$-8\text{Tr}\Big(M_F\left(M_F\right)^*\left(y_2\right)^T\left(y_2\right)^*\Big)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{27g_1^2m_2^2}{2}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$	$-12\text{Tr}\Big(M_F\left(M_F\right)^*\left(y_2\right)^T\left(y_2\right)^*\Big)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{27g_1^2m_2^2}{2}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: m_2^2

Terms 4 of 5

Column	Representative term structure		
T1	$-8\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\Big)$		
T2	$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$		
T3	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-8\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\Big)$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-12\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\Big)$	$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: m_2^2

Terms 5 of 5

Column	Representative term structure
T1	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, $-\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, $-\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, $-\frac{3}{2}$)	(2, $-\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, $-\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, $-\frac{3}{2}$)	(2, $-\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, $-\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, $-\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, $-\frac{3}{2}$)	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, $-\frac{1}{2}$)	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Quantity: λ_1

Terms 1 of 7

[illegible]

Quantity: λ_1

Terms 2 of 7

Column	Representative term structure
T1	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$
T2	$+4\lambda_5\left(\lambda_5\right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$

Quantity: λ_1

Terms 3 of 7

Column		Representative term structure
T1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	
T2	$-9g_2^2\lambda_1$	
T3	$+\frac{3g_1^4}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-E-p2	2	$(3, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Quantity: λ_1

Terms 4 of 7

Column	Representative term structure
T1	$+12\lambda_1^2$
T2	$+2\lambda_{HS_1}^{(1)2}$
T3	$+6\lambda_{HS_2}^{(1)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$

Quantity: λ_1

Terms 5 of 7

Column		Representative term structure
T1	$+3\lambda_{HS_2}^{(A)^2}$	
T2	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	
T3	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	—	—

Quantity: λ_1

Terms 6 of 7

Column		Representative term structure
T1		$+8\lambda_7\left(\lambda_7\right)^*$
T2		$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3		$+8\lambda_8\left(\lambda_8\right)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_7\left(\lambda_7\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+8\lambda_8\left(\lambda_8\right)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_7\left(\lambda_7\right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+8\lambda_8\left(\lambda_8\right)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_1

Terms 7 of 7

Column	Representative term structure
T1	$+3\lambda_{HS_1}^{(A)^2}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+3\lambda_{HS_1}^{(A)^2}$

$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(1)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

Terms 1 of 9

Column		Representative term structure
T1	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	
T2	$-8\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	
T3	$+192g_1^4$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-8\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	$+192g_1^4$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-8\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	$+12g_1^4$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-8\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-4\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-4\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-4\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$6\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-5\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$6\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-5\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$6\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-5\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-3\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	$+12g_1^4$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-3\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-3\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	$+12g_1^4$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$-3\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	$+192g_1^4$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$-3\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	$+12g_1^4$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$-3\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$-3\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$	$+12g_1^4$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 9

Column	Representative term structure
T1	$-48g_1^2\lambda_{S_1}^{(1)}$
T2	$+6\lambda_{12}^{(1)2}$
T3	$+4\lambda_{HS_1}^{(1)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-48g_1^2\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-12g_1^2\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 9

Column		Representative term structure
T1	$+10\lambda_{S_1}^{(1)2}$	
T2	$+\frac{9g_2^4}{4}$	
T3	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_1}^{(1)}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+10\lambda_{S_1}^{(1)2}$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+10\lambda_{S_1}^{(1)2}$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+10\lambda_{S_1}^{(1)2}$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+12\lambda_{S_1}^{(1)2}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_1}^{(1)}\right)$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+12\lambda_{S_1}^{(1)2}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+12\lambda_{S_1}^{(1)2}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+12\lambda_{S_1}^{(1)2}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_1}^{(1)}\right)$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+12\lambda_{S_1}^{(1)2}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+12\lambda_{S_1}^{(1)2}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+14\lambda_{S_1}^{(1)2}$	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 9

Column		Representative term structure
T1	$-9g_2^2\lambda_{S_1}^{(1)}$	
T2	$+\frac{243g_1^4}{4}$	
T3	$+\frac{\lambda_{12}^{(A)2}}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{243g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{243g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 5 of 9

Column	Representative term structure	
T1	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	
T2	$+\lambda_{11}(2\operatorname{Tr}\big(y_1\ (y_2)^\dagger\big)+24\,(\lambda_{11})^*)$	
T3	$+2\,(\lambda_{11})^*\operatorname{Tr}\big(y_2\ (y_1)^\dagger\big)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	$+\lambda_{11}(2\operatorname{Tr}\big(y_1\ (y_2)^\dagger\big)+24\,(\lambda_{11})^*)$	$+2\,(\lambda_{11})^*\operatorname{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	$+\lambda_{11}(3\operatorname{Tr}\big(y_1\ (y_2)^\dagger\big)+24\,(\lambda_{11})^*)$	$+3\,(\lambda_{11})^*\operatorname{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 6 of 9

Column	Representative term structure
T1	$+8\lambda_7\left(\lambda_7\right)^*$
T2	$+8\lambda_9\left(\lambda_9\right)^*$
T3	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$+8\lambda_7\left(\lambda_7\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$+8\lambda_7\left(\lambda_7\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	—	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)}+14\lambda_{S_1}^{(A)})$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 7 of 9

Column	Representative term structure	
T1	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	
T2	$+21g_2^4$	
T3	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$	$+g_1^2\left(48g_2^2-48\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)}\right)$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 8 of 9

Column	Representative term structure
T1	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$
T2	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3	$+26\lambda_{S_1}^{(A)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$	$+26\lambda_{S_1}^{(A)2}$
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$	$+26\lambda_{S_1}^{(A)2}$
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$	$+26\lambda_{S_1}^{(A)2}$
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$	$+26\lambda_{S_1}^{(A)2}$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$	$+26\lambda_{S_1}^{(A)2}$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$	$+26\lambda_{S_1}^{(A)2}$
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$	$+26\lambda_{S_1}^{(A)2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 9 of 9

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T2	$+6\lambda_{12}^{(A)2}$
T3	$+6\lambda_{12}^{(\times)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$

$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}^{(1)}}^{(1)}$$

Terms 1 of 9

Column		Representative term structure
T1	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	
T2	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	
T3	$-3\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	—	$-4\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	—	—	$-4\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	—	$-4\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	—	$-5\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	—	—	$-5\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	—	$-5\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	—	—	$-8\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	—	—	$-8\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	—	—	$-8\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\operatorname{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 9

Column	Representative term structure		
T1	$+21g_2^4$		
T2	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$		
T3	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+21g_2^4$	—	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	$+21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+21g_2^4$	—	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$+21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$+21g_2^4$	$+g_1^2\left(48g_2^2-48\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 9

Column	Representative term structure
T1	$+12g_1^4$
T2	$+2\lambda_{12}^{(1)2}$
T3	$+4\lambda_{HS_2}^{(1)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+12g_1^4$	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+12g_1^4$	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$+12g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$+192g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+12g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+12g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$+192g_1^4$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 9

Column		Representative term structure
T1		$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$
T2		$+14\lambda_{S_2}^{(1)^2}$
T3		$+26\lambda_{S_2}^{(A)^2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+26\lambda_{S_2}^{(A)^2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+26\lambda_{S_2}^{(A)^2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+26\lambda_{S_2}^{(A)^2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+12\lambda_{S_2}^{(1)^2}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$+10\lambda_{S_2}^{(1)^2}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$+10\lambda_{S_2}^{(1)^2}$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$+10\lambda_{S_2}^{(1)^2}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+26\lambda_{S_2}^{(A)^2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+26\lambda_{S_2}^{(A)^2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+26\lambda_{S_2}^{(A)^2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+26\lambda_{S_2}^{(A)^2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 5 of 9

Column	Representative term structure		
T1	$4\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$		
T2	$+\frac{9g_2^4}{4}$		
T3	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$4\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+4\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$4\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$6\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+6\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$6\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$8\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$8\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$8\lambda_{S_2}^{(1)}T_{\nu}^{(2)}$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 6 of 9

Column		Representative term structure
T1	$-9g_2^2\lambda_{S_2}^{(1)}$	
T2	$+\frac{3g_1^4}{4}$	
T3	$+\frac{\lambda_{12}^{(A)2}}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{243g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{3g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-9g_2^2\lambda_{S_2}^{(1)}$	$+\frac{243g_1^4}{4}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 7 of 9

Column	Representative term structure	
T1	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	
T2	$\lambda_{10}(2\operatorname{Tr}\left(y_1\ (y_2)^{\dagger}\right)+24\left(\lambda_{10}\right)^{*})$	
T3	$+2\left(\lambda_{10}\right)^{*}\operatorname{Tr}\left(y_2\ (y_1)^{\dagger}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$\lambda_{10}(2\operatorname{Tr}\left(y_1\ (y_2)^{\dagger}\right)+24\left(\lambda_{10}\right)^{*})$	$+2\left(\lambda_{10}\right)^{*}\operatorname{Tr}\left(y_2\ (y_1)^{\dagger}\right)$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$\lambda_{10}(3\operatorname{Tr}\left(y_1\ (y_2)^{\dagger}\right)+24\left(\lambda_{10}\right)^{*})$	$+3\left(\lambda_{10}\right)^{*}\operatorname{Tr}\left(y_2\ (y_1)^{\dagger}\right)$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 8 of 9

Column	Representative term structure	
T1	$+8\lambda_8\left(\lambda_8\right)^*$	
T2	$+8\lambda_9\left(\lambda_9\right)^*$	
T3	$-12g_1^2\lambda_{S_2}^{(1)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_8\left(\lambda_8\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_8\left(\lambda_8\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$-12g_1^2\lambda_{S_2}^{(1)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$-48g_1^2\lambda_{S_2}^{(1)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 9 of 9

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T2	$+6\lambda_{12}^{(A)2}$
T3	$+6\lambda_{12}^{(\times)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$

$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Terms 1 of 9

Column		Representative term structure
T1	$\lambda_{HS_1}^{(1)}$	$(4T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$-4\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	
T3	$+12\lambda_5\,(\lambda_5)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-4\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-4\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-4\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	$-3\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	$-3\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	$-3\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	$-2\operatorname{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 9

Column		Representative term structure
T1		$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$
T2		$+12g_1^4$
T3		$-\frac{51g_1^2\lambda_{HS_1}^{(1)}}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+12g_1^4$	$-\frac{51g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	$-$	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-$	$-$	$-$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-$	$-$	$-$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-$	$-$	$-$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-$	$-$	$-$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-$	$-$	$-$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-$	$-$	$-$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$-$	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+12g_1^4$	$-\frac{51g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$-$	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 9

Column		Representative term structure
T1		$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T2		$+6\lambda_1\lambda_{HS_1}^{(1)}$
T3		$+4\lambda_{HS_1}^{(1)^2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 4 of 9

Column		Representative term structure
T1		$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T2		$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T3		$+\frac{9g_2^4}{4}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{9g_2^4}{4}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{9g_2^4}{4}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{9g_2^4}{4}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	$+\frac{9g_2^4}{4}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	$+\frac{9g_2^4}{4}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	$+\frac{9g_2^4}{4}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 6 of 9

Column		Representative term structure
T1		$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$
T2		$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*\right)$
T3		$+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*\right)$	$+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*\right)$	$+\frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 7 of 9

Column	Representative term structure
T1	$+24\lambda_7\left(\lambda_7\right)^*$
T2	$+3\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3	$\lambda_{HS_1}^{(1)}\left(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+24\lambda_7\left(\lambda_7\right)^*$	$+3\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	$\lambda_{HS_1}^{(1)}\left(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+24\lambda_7\left(\lambda_7\right)^*$	$+3\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$\lambda_{HS_1}^{(1)}\left(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	$\lambda_{HS_1}^{(1)}\left(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 8 of 9

Column		Representative term structure
T1	$\lambda_{HS_1}^{(1)}$	$(2T_\nu^{(1)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda_{S_1}^{(A)})$
T2	$+6g_2^4$	
T3	$+6\lambda_{HS_1}^{(A)^2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)^2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 9 of 9

Column	Representative term structure
T1	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_2}^{(1)}$

$$16\pi^2\,\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

Terms 1 of 9

Column		Representative term structure
T1	$\lambda_{HS_2}^{(1)}$	$(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$
T2	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	
T3	$+4\lambda_5\,(\lambda_5)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-3\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-3\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-3\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 2 of 9

Column	Representative term structure
T1	$+6g_2^4$
T2	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$
T3	$+3g_1^4$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+12g_1^4$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+12g_1^4$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 9

Column		Representative term structure
T1	-	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T2	+	$2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$
T3	+	$6\lambda_1\lambda_{HS_2}^{(1)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-\frac{51g_1^2\lambda_{HS_2}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$-\frac{51g_1^2\lambda_{HS_2}^{(1)}}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 9

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1) \ 2}$
T2	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T3	$+6\lambda_{HS_2}^{(A) \ 2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A) \ 2}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A) \ 2}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A) \ 2}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	$+4\lambda_{HS_2}^{(1) \ 2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+4\lambda_{HS_2}^{(1) \ 2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	$+4\lambda_{HS_2}^{(1) \ 2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$+4\lambda_{HS_2}^{(1) \ 2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+4\lambda_{HS_2}^{(1) \ 2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$+4\lambda_{HS_2}^{(1) \ 2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A) \ 2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A) \ 2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A) \ 2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A) \ 2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 5 of 9

Column	Representative term structure
T1	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+ \frac{9g_2^4}{4}$
T3	$-9g_2^2\lambda_{HS_2}^{(1)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 6 of 9

Column		Representative term structure
T1	$+ \frac{3g_1^4}{4}$	
T2	$-3g_1^2\lambda_{HS_2}^{(1)}$	
T3	$+ \frac{3\lambda_{HS_2}^{(A)2}}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+ \frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_2}^{(1)}$	$+ \frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+ \frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_2}^{(1)}$	$+ \frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+ \frac{27g_1^4}{4}$	$-15g_1^2\lambda_{HS_2}^{(1)}$	$+ \frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+ \frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_2}^{(1)}$	$+ \frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+ \frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_2}^{(1)}$	$+ \frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+ \frac{27g_1^4}{4}$	$-15g_1^2\lambda_{HS_2}^{(1)}$	$+ \frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 7 of 9

Column		Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*\right)$	
T2	$+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	
T3	$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*\right)$	$+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*\right)$	$+\frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 8 of 9

Column	Representative term structure
T1	$+24\lambda_8\left(\lambda_8\right)^*$
T2	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+24\lambda_8\left(\lambda_8\right)^*$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+24\lambda_8\left(\lambda_8\right)^*$	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 9 of 9

Column	Representative term structure									
T1	$+ \lambda_{HS_1}^{(1)} (6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)})$									

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+ \lambda_{HS_1}^{(1)} (6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)})$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+ \lambda_{HS_1}^{(1)} (6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)})$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+ \lambda_{HS_1}^{(1)} (6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)})$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+ \lambda_{HS_1}^{(1)} (6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)})$

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

$$16\pi^2\,\mu\,\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$$

Terms 1 of 12

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$
T2	$-4\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$
T3	$+4\lambda_5\,(\lambda_5)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$-2\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	$+3\lambda_5\,(\lambda_5)^*$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$-2\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	$+3\lambda_5\,(\lambda_5)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$-2\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	$+3\lambda_5\,(\lambda_5)^*$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	$+3\lambda_5\,(\lambda_5)^*$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	$+3\lambda_5\,(\lambda_5)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	$+3\lambda_5\,(\lambda_5)^*$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-4\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-4\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-4\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$-2\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$-2\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$-2\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$-2\,\text{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_2\,(y_2)^{\dagger}\Big)$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 12

Column	Representative term structure	
T1	$-12g_2^2\lambda_{12}^{(1)}$	
T2	$+48g_1^4$	
T3	$-30g_1^2\lambda_{12}^{(1)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$-12g_2^2\lambda_{12}^{(1)}$	$+48g_1^4$	$-30g_1^2\lambda_{12}^{(1)}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$-12g_2^2\lambda_{12}^{(1)}$	—	$-6g_1^2\lambda_{12}^{(1)}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$-12g_2^2\lambda_{12}^{(1)}$	—	$-6g_1^2\lambda_{12}^{(1)}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-9g_2^2\lambda_{12}^{(1)}$	—	$-15g_1^2\lambda_{12}^{(1)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-9g_2^2\lambda_{12}^{(1)}$	—	$-3g_1^2\lambda_{12}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-9g_2^2\lambda_{12}^{(1)}$	—	$-15g_1^2\lambda_{12}^{(1)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-9g_2^2\lambda_{12}^{(1)}$	—	$-15g_1^2\lambda_{12}^{(1)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-9g_2^2\lambda_{12}^{(1)}$	—	$-3g_1^2\lambda_{12}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-9g_2^2\lambda_{12}^{(1)}$	—	$-15g_1^2\lambda_{12}^{(1)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$-12g_2^2\lambda_{12}^{(1)}$	—	$-6g_1^2\lambda_{12}^{(1)}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$-12g_2^2\lambda_{12}^{(1)}$	—	$-6g_1^2\lambda_{12}^{(1)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$-12g_2^2\lambda_{12}^{(1)}$	$+48g_1^4$	$-30g_1^2\lambda_{12}^{(1)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$+48g_1^4$	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$+48g_1^4$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 12

Column	Representative term structure		
T1	$+4\lambda_{12}^{(1)2}$		
T2	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$		
T3	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{12}^{(1)2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{12}^{(1)2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{12}^{(1)2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+4\lambda_{12}^{(1)2}$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{12}^{(1)2}$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{12}^{(1)2}$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{12}^{(1)2}$	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 12

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T2	$\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$
T3	$+\frac{9g_2^4}{4}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	$\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$+\frac{9g_2^4}{4}$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	$\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$+\frac{9g_2^4}{4}$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	$\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$+\frac{9g_2^4}{4}$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	$+\frac{9g_2^4}{4}$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	$+\frac{9g_2^4}{4}$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	$+\frac{9g_2^4}{4}$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 12

Column		Representative term structure
T1		$+ \frac{27 g_1^4}{4}$
T2		$+ \frac{3 \lambda_{12}^{(A)2}}{4}$
T3		$+ \lambda_{10} \left(\frac{3}{2} \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + 9 ((\lambda_{10})^* + (\lambda_{11})^*) \right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+ \frac{27 g_1^4}{4}$	$+ \frac{3 \lambda_{12}^{(A)2}}{4}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+ \frac{3 g_1^4}{4}$	$+ \frac{3 \lambda_{12}^{(A)2}}{4}$	$+ \lambda_{10} \left(\frac{3}{2} \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + 9 ((\lambda_{10})^* + (\lambda_{11})^*) \right)$ $+ \lambda_{11} \left(\frac{3}{2} \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + 9 ((\lambda_{10})^* + (\lambda_{11})^*) \right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+ \frac{27 g_1^4}{4}$	$+ \frac{3 \lambda_{12}^{(A)2}}{4}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+ \frac{27 g_1^4}{4}$	$+ \frac{3 \lambda_{12}^{(A)2}}{4}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+ \frac{3 g_1^4}{4}$	$+ \frac{3 \lambda_{12}^{(A)2}}{4}$	$+ \lambda_{10} \left(\frac{9}{4} \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + 9 ((\lambda_{10})^* + (\lambda_{11})^*) \right)$ $+ \lambda_{11} \left(\frac{9}{4} \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + 9 ((\lambda_{10})^* + (\lambda_{11})^*) \right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+ \frac{27 g_1^4}{4}$	$+ \frac{3 \lambda_{12}^{(A)2}}{4}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 6 of 12

Column	Representative term structure
T1	$+\frac{3}{2}((\lambda_{10})^*+(\lambda_{11})^*)\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$
T2	$-4\text{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$
T3	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+\frac{3}{2}((\lambda_{10})^*+(\lambda_{11})^*)\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$	$-4\text{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+\frac{9}{4}((\lambda_{10})^*+(\lambda_{11})^*)\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$	$-3\text{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 7 of 12

Column	Representative term structure
T1	$+24\lambda_9\left(\lambda_9\right)^*$
T2	$\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$
T3	$-\frac{9}{2}\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\Big)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$+24\lambda_9\left(\lambda_9\right)^*$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	$\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$	$-\frac{9}{2}\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\Big)$
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$+24\lambda_9\left(\lambda_9\right)^*$	$\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$	$-\frac{9}{2}\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\Big)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	$\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$	$-\frac{9}{2}\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\Big)$
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 8 of 12

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)} + 4T_{\nu}^{(2)} + 4\lambda_{S_1}^{(A)})$
T2	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)} + 2T_{\nu}^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$
T3	$+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)} + \frac{4}{3}T_{\nu}^{(2)}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)} + 4T_{\nu}^{(2)} + 4\lambda_{S_1}^{(A)})$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)} + 4T_{\nu}^{(2)} + 4\lambda_{S_1}^{(A)})$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)} + 4T_{\nu}^{(2)} + 4\lambda_{S_1}^{(A)})$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)} + 2T_{\nu}^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)} + \frac{4}{3}T_{\nu}^{(2)}\right)$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)} + 2T_{\nu}^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)} + \frac{4}{3}T_{\nu}^{(2)}\right)$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)} + 2T_{\nu}^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)} + \frac{4}{3}T_{\nu}^{(2)}\right)$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)} + 2T_{\nu}^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)} + \frac{4}{3}T_{\nu}^{(2)}\right)$

Quantity: $\lambda_{12}^{(1)}$

Terms 9 of 12

Column	Representative term structure
T1	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$
T2	$+16g_2^4$
T3	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$

Quantity: $\lambda_{12}^{(1)}$

Terms 10 of 12

Column		Representative term structure
T1		$+g_1^2(-30\lambda_{12}^{(1)}-20\lambda_{12}^{(\times)})$
T2		$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3		$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	(1,0)	(3,1)	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1,-1)	—	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1,0)	—	—	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1,1)	—	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3,-1)	—	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3,0)	—	—	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	$(2,-\frac{1}{2})$	—	—	—
T3-D-p0	0	(3,0)	(1,1)	$(2,\frac{1}{2})$	—	—	—
T3-D-p2	2	(3,1)	(1,2)	$(2,\frac{3}{2})$	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	$(2,-\frac{3}{2})+g_1^2(-30\lambda_{12}^{(1)}-20\lambda_{12}^{(\times)})$		$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E-m2	-2	(3,-1)	(3,0)	$(2,-\frac{1}{2})+g_1^2(-6\lambda_{12}^{(1)}-4\lambda_{12}^{(\times)})$		$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E-p0	0	(3,0)	(3,1)	$(2,\frac{1}{2})+g_1^2(-6\lambda_{12}^{(1)}-4\lambda_{12}^{(\times)})$		$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E-p2	2	(3,1)	(3,2)	$(2,\frac{3}{2})+g_1^2(-30\lambda_{12}^{(1)}-20\lambda_{12}^{(\times)})$		$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$

Quantity: $\lambda_{12}^{(1)}$

Terms 11 of 12

Column		Representative term structure
T1		$+16\lambda_{12}^{(A)2}$
T2		$+\frac{32\lambda_{12}^{(\times)2}}{3}$
T3		$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+16\lambda_{12}^{(A)2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+16\lambda_{12}^{(A)2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+16\lambda_{12}^{(A)2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+16\lambda_{12}^{(A)2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$

Quantity: $\lambda_{12}^{(1)}$

Terms 12 of 12

Column		Representative term structure
T1		$+ \frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+ \frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+ \frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+ \frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+ \frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

$\beta_{\lambda_5}^{(1)}$

Quantity: λ_5

$$16\pi^2\mu\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 6

Column		Representative term structure
T1	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{\overline{HS}_2}^{(A)})$	
T2	$-\frac{21g_2^2\lambda_5}{2}$	
T3	$-\frac{33g_1^2\lambda_5}{2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{\overline{HS}_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{\overline{HS}_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{\overline{HS}_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$

Quantity: λ_5

Terms 2 of 6

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Quantity: λ_5

Terms 3 of 6

Column		Representative term structure
T1		$+4\lambda_{HS_2}^{(1)}\lambda_5$
T2		$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$
T3		$-9g_2^2\lambda_5$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_5$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_5$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_5$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	$-9g_2^2\lambda_5$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	$-9g_2^2\lambda_5$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	$-9g_2^2\lambda_5$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—

Quantity: λ_5

Terms 4 of 6

Column	Representative term structure	
T1	$-9g_1^2\lambda_5$	
T2	$+2\left(\lambda_7\right)^*\mathrm{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)$	
T3	$+2\left(\lambda_8\right)^*\mathrm{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$-9g_1^2\lambda_5$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-3g_1^2\lambda_5$	$+2\left(\lambda_7\right)^*\mathrm{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)$	$+2\left(\lambda_8\right)^*\mathrm{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$-9g_1^2\lambda_5$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$-9g_1^2\lambda_5$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-3g_1^2\lambda_5$	$+3\left(\lambda_7\right)^*\mathrm{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)$	$+3\left(\lambda_8\right)^*\mathrm{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$-9g_1^2\lambda_5$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_5

Terms 5 of 6

Column	Representative term structure
T1	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_7)^*$
T2	$+4\lambda_{11}(\lambda_7)^*$
T3	$+4(\lambda_8)^*((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_7)^*$	$+4\lambda_{11}(\lambda_7)^*$	$+4(\lambda_8)^*((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_7)^*$	$+4\lambda_{11}(\lambda_7)^*$	$+4(\lambda_8)^*((\lambda_{10})^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Terms 6 of 6

Column		Representative term structure
T1	λ_5	$\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$
T2	λ_5	$(T_\nu^{(1)}+2T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS1}^{(A)})$
T3	λ_5	$\left(T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS1}^{(A)}-\frac{4\lambda_{HS2}^{(A)}}{3}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\lambda_5\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\lambda_5\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\lambda_5\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS1}^{(A)})$	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS1}^{(A)})$	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS1}^{(A)})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS1}^{(A)}-\frac{4\lambda_{HS2}^{(A)}}{3}\right)$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS1}^{(A)})$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS1}^{(A)})$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS1}^{(A)})$

$\beta_{\lambda_{12}^{(A)}}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{(A)}}{d\mu}=\beta_{\lambda_{12}^{(A)}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda_{12}^{(A)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$
T2	$-\frac{8}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$
T3	$+\frac{4}{3}\lambda_5\ (\lambda_5)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\lambda_{12}^{(A)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$-\frac{8}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$+\frac{4}{3}\lambda_5\ (\lambda_5)^*$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_{12}^{(A)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$-\frac{8}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$+\frac{4}{3}\lambda_5\ (\lambda_5)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\lambda_{12}^{(A)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$-\frac{8}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$+\frac{4}{3}\lambda_5\ (\lambda_5)^*$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{2}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$+\frac{4}{3}\lambda_5\ (\lambda_5)^*$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{2}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$+\frac{4}{3}\lambda_5\ (\lambda_5)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{2}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$+\frac{4}{3}\lambda_5\ (\lambda_5)^*$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$+\frac{2}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$-\frac{2}{9}\lambda_5\ (\lambda_5)^*$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+\frac{2}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$-\frac{2}{9}\lambda_5\ (\lambda_5)^*$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$+\frac{2}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$-\frac{2}{9}\lambda_5\ (\lambda_5)^*$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$+\frac{2}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	$-\frac{2}{9}\lambda_5\ (\lambda_5)^*$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 8

Column	Representative term structure	
T1	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	
T2	$-9g_2^2\lambda_{12}^{(A)}$	
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	$-9g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	$-9g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+g_1^2\left(16g_2^2-30\lambda_{12}^{(A)}\right)$	$-24g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	$-24g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	$-24g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$+g_1^2\left(16g_2^2-30\lambda_{12}^{(A)}\right)$	$-24g_2^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 8

Column	Representative term structure	
T1	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	
T2	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	
T3	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—

Quantity: $\lambda_{12}^{(A)}$

Terms 4 of 8

Column	Representative term structure	
T1	$+ \lambda_{10} \left(\frac{2}{3} \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} (5 \left(\lambda_{10} \right)^* + \left(\lambda_{11} \right)^*) \right)$	
T2	$+ \lambda_{11} \left(\frac{2}{3} \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} ((\lambda_{10})^* + 5 \left(\lambda_{11} \right)^*) \right)$	
T3	$+ \frac{2}{3} ((\lambda_{10})^* + (\lambda_{11})^*) \operatorname{Tr} \left(y_2 \left(y_1 \right)^{\dagger} \right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+ \lambda_{10} \left(\frac{2}{3} \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} (5 \left(\lambda_{10} \right)^* + \left(\lambda_{11} \right)^*) \right)$	$+ \lambda_{11} \left(\frac{2}{3} \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} ((\lambda_{10})^* + 5 \left(\lambda_{11} \right)^*) \right)$	$+ \frac{2}{3} ((\lambda_{10})^* + (\lambda_{11})^*) \operatorname{Tr} \left(y_2 \left(y_1 \right)^{\dagger} \right)$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 5 of 8

Column	Representative term structure	
T1	$+\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$	
T2	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	
T3	$\lambda_{12}^{(A)}\left(3T_\nu^{(1)}+3T_\nu^{(2)}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$\left(2,-\frac{3}{2}\right)$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$\left(2,-\frac{1}{2}\right)$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$\left(2,\frac{1}{2}\right)$	—	—	—
T3-B-m3	-3	$\left(2,-\frac{3}{2}\right)$	$\left(2,-\frac{1}{2}\right)$	(1, -1)	—	—	—
T3-B-m1	-1	$\left(2,-\frac{1}{2}\right)$	$\left(2,\frac{1}{2}\right)$	(1, 0)	$+\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	—
T3-B-p1	1	$\left(2,\frac{1}{2}\right)$	$\left(2,\frac{3}{2}\right)$	(1, 1)	—	—	—
T3-C-m3	-3	$\left(2,-\frac{3}{2}\right)$	$\left(2,-\frac{1}{2}\right)$	(3, -1)	—	—	$\lambda_{12}^{(A)}\left(3T_\nu^{(1)}+3T_\nu^{(2)}\right)$
T3-C-m1	-1	$\left(2,-\frac{1}{2}\right)$	$\left(2,\frac{1}{2}\right)$	(3, 0)	$+\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	$\lambda_{12}^{(A)}\left(3T_\nu^{(1)}+3T_\nu^{(2)}\right)$
T3-C-p1	1	$\left(2,\frac{1}{2}\right)$	$\left(2,\frac{3}{2}\right)$	(3, 1)	—	—	$\lambda_{12}^{(A)}\left(3T_\nu^{(1)}+3T_\nu^{(2)}\right)$
T3-D-m2	-2	(3, -1)	(1, 0)	$\left(2,-\frac{1}{2}\right)$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$\left(2,\frac{1}{2}\right)$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$\left(2,\frac{3}{2}\right)$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$\left(2,-\frac{3}{2}\right)$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$\left(2,-\frac{1}{2}\right)$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$\left(2,\frac{1}{2}\right)$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$\left(2,\frac{3}{2}\right)$	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 6 of 8

Column	Representative term structure
T1	$+ \lambda_{10} \left(\text{Tr} \left(y_1 \ (y_2)^\dagger \right) + \frac{4}{3} (5 \ (\lambda_{10})^* + (\lambda_{11})^*) \right)$
T2	$+ \lambda_{11} \left(\text{Tr} \left(y_1 \ (y_2)^\dagger \right) + \frac{4}{3} ((\lambda_{10})^* + 5 \ (\lambda_{11})^*) \right)$
T3	$+ ((\lambda_{10})^* + (\lambda_{11})^*) \text{Tr} \left(y_2 \ (y_1)^\dagger \right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+ \lambda_{10} \left(\text{Tr} \left(y_1 \ (y_2)^\dagger \right) + \frac{4}{3} (5 \ (\lambda_{10})^* + (\lambda_{11})^*) \right)$	$+ \lambda_{11} \left(\text{Tr} \left(y_1 \ (y_2)^\dagger \right) + \frac{4}{3} ((\lambda_{10})^* + 5 \ (\lambda_{11})^*) \right)$	$+ ((\lambda_{10})^* + (\lambda_{11})^*) \text{Tr} \left(y_2 \ (y_1)^\dagger \right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 7 of 8

Column		Representative term structure
T1	$-\frac{8}{3}\text{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$	
T2	$\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	
T3	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{8}{3}\text{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 8 of 8

Column	Representative term structure	
T1	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	
T2	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	
T3	$-6g_1{}^2\lambda_{12}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	$-6g_1{}^2\lambda_{12}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	$-6g_1{}^2\lambda_{12}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	—

$\beta_{\lambda_{12}^{(\times)}}^{(1)}$

Quantity: $\lambda_{12}^{(\times)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{(\times)}}{d\mu}=\beta_{\lambda_{12}^{(\times)}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure			
T1	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+\frac{1}{2}T_\nu^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$			
T2	$+\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$			
T3	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$			

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+\frac{1}{2}T_\nu^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+\frac{1}{2}T_\nu^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+\frac{1}{2}T_\nu^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+\frac{1}{2}T_\nu^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$

Quantity: $\lambda_{12}^{(\times)}$

Terms 2 of 6

Column	Representative term structure	
T1	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	
T2	$+\frac{33g_2^4}{4}$	
T3	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$\left(2,-\frac{3}{2}\right)$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$\left(2,-\frac{1}{2}\right)$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$\left(2,\frac{1}{2}\right)$	—	—	—
T3-B-m3	-3	$\left(2,-\frac{3}{2}\right)$	$\left(2,-\frac{1}{2}\right)$	(1, -1)	—	—	—
T3-B-m1	-1	$\left(2,-\frac{1}{2}\right)$	$\left(2,\frac{1}{2}\right)$	(1, 0)	—	—	—
T3-B-p1	1	$\left(2,\frac{1}{2}\right)$	$\left(2,\frac{3}{2}\right)$	(1, 1)	—	—	—
T3-C-m3	-3	$\left(2,-\frac{3}{2}\right)$	$\left(2,-\frac{1}{2}\right)$	(3, -1)	—	—	—
T3-C-m1	-1	$\left(2,-\frac{1}{2}\right)$	$\left(2,\frac{1}{2}\right)$	(3, 0)	—	—	—
T3-C-p1	1	$\left(2,\frac{1}{2}\right)$	$\left(2,\frac{3}{2}\right)$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$\left(2,-\frac{1}{2}\right)$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$\left(2,\frac{1}{2}\right)$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$\left(2,\frac{3}{2}\right)$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$\left(2,-\frac{3}{2}\right)$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$
T3-E-m2	-2	(3, -1)	(3, 0)	$\left(2,-\frac{1}{2}\right)$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$
T3-E-p0	0	(3, 0)	(3, 1)	$\left(2,\frac{1}{2}\right)$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$
T3-E-p2	2	(3, 1)	(3, 2)	$\left(2,\frac{3}{2}\right)$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$

Quantity: $\lambda_{12}^{(\times)}$

Terms 3 of 6

Column	Representative term structure	
T1	$+12g_1^4$	
T2	$+g_1^2\left(-\frac{15\lambda_{12}^{(1)}}{2}-30\lambda_{12}^{(\times)}\right)$	
T3	$+\lambda_{12}^{(1)2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+12g_1^4$	$+g_1^2\left(-\frac{15\lambda_{12}^{(1)}}{2}-30\lambda_{12}^{(\times)}\right)$	$+\lambda_{12}^{(1)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	$+g_1^2\left(-\frac{3\lambda_{12}^{(1)}}{2}-6\lambda_{12}^{(\times)}\right)$	$+\lambda_{12}^{(1)2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	$+g_1^2\left(-\frac{3\lambda_{12}^{(1)}}{2}-6\lambda_{12}^{(\times)}\right)$	$+\lambda_{12}^{(1)2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+12g_1^4$	$+g_1^2\left(-\frac{15\lambda_{12}^{(1)}}{2}-30\lambda_{12}^{(\times)}\right)$	$+\lambda_{12}^{(1)2}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 4 of 6

Column	Representative term structure
T1	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$
T2	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$
T3	$+4\lambda_{12}^{(A)2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 5 of 6

Column	Representative term structure	
T1	$+11\lambda_{12}^{(\times)2}$	
T2	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	
T3	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+11\lambda_{12}^{(\times)2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+11\lambda_{12}^{(\times)2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+11\lambda_{12}^{(\times)2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+11\lambda_{12}^{(\times)2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 6 of 6

Column	Representative term structure	
T1	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$$\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_1}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{HS_1}^{(A)}}{d\mu}=\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Terms 1 of 8

Column		Representative term structure
T1	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$	
T2	$+\frac{8}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	
T3	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$	$+\frac{8}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$	$+\frac{8}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$	$+\frac{8}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$+\frac{16}{3}\lambda_5\,(\lambda_5)^*$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$-\frac{8}{3}\lambda_5\,(\lambda_5)^*$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$-\frac{8}{3}\lambda_5\,(\lambda_5)^*$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$-\frac{8}{3}\lambda_5\,(\lambda_5)^*$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$-\frac{4}{3}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 8

Column	Representative term structure
T1	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T2	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T3	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 4 of 8

Column		Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*\right)\right)$	
T2	$-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	
T3	$+\frac{64}{3}\lambda_7\left(\lambda_7\right)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*\right)\right)$	$-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+\frac{64}{3}\lambda_7\left(\lambda_7\right)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	$+\frac{64}{3}\lambda_7\left(\lambda_7\right)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 5 of 8

Column		Representative term structure
T1	$-\frac{4}{3}\lambda_{11}$	$\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$+g_1^2$	$\left(6g_2^2-3\lambda_{HS_1}^{(A)}\right)$
T3	$\lambda_{HS_1}^{(A)}$	$\left(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$+g_1^2\left(6g_2^2-3\lambda_{HS_1}^{(A)}\right)$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	$\lambda_{HS_1}^{(A)}\left(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	—	$\lambda_{HS_1}^{(A)}\left(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	$+g_1^2\left(6g_2^2-3\lambda_{HS_1}^{(A)}\right)$	$\lambda_{HS_1}^{(A)}\left(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 6 of 8

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\text{Tr}\left(y_1\ (y_2)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*\right)\right)$
T2	$-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda_{S_1}^{(A)})$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\text{Tr}\left(y_1\ (y_2)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*\right)\right)$	$-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda_{S_1}^{(A)})$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda_{S_1}^{(A)})$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda_{S_1}^{(A)})$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda_{S_1}^{(A)})$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda_{S_1}^{(A)})$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda_{S_1}^{(A)})$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda_{S_1}^{(A)})$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 7 of 8

Column		Representative term structure
T1	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	
T2	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	
T3	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$+g_1^2\left(-8g_2^2-\frac{51\lambda_{HS_1}^{(A)}}{2}\right)$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 8 of 8

Column		Representative term structure
T1	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	
T2	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

$$\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_2}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{HS_2}^{(A)}}{d\mu}=\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Terms 1 of 8

Column		Representative term structure
T1	$\lambda_{HS_2}^{(A)}$	$(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$
T2	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	
T3	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_5\,(\lambda_5)^*$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$-\frac{8}{9}\lambda_5\,(\lambda_5)^*$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 8

Column	Representative term structure	
T1	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	
T2	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	
T3	$+2\lambda_1\lambda_{HS_2}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 8

Column		Representative term structure
T1		$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T2		$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
T3		$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 4 of 8

Column	Representative term structure
T1	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T2	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3	$+8\operatorname{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+8\operatorname{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+8\operatorname{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+8\operatorname{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	$-4\operatorname{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	$-4\operatorname{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	$-4\operatorname{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+g_1^2\left(8g_2^2-\frac{51\lambda_{HS_2}^{(A)}}{2}\right)$	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 5 of 8

Column	Representative term structure	
T1	$+16\lambda_5\left(\lambda_5\right)^*$	
T2	$+g_1^2\left(-6g_2^2-3\lambda_{HS_2}^{(A)}\right)$	
T3	$-9g_2^2\lambda_{HS_2}^{(A)}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	$+16\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_{HS_2}^{(A)}$
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+16\lambda_5\left(\lambda_5\right)^*$	—	$-9g_2^2\lambda_{HS_2}^{(A)}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	$+16\lambda_5\left(\lambda_5\right)^*$	—	$-9g_2^2\lambda_{HS_2}^{(A)}$
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	$+16\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_{HS_2}^{(A)}$
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+16\lambda_5\left(\lambda_5\right)^*$	—	$-9g_2^2\lambda_{HS_2}^{(A)}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	$+16\lambda_5\left(\lambda_5\right)^*$	—	$-9g_2^2\lambda_{HS_2}^{(A)}$
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 6 of 8

Column Representative term structure	
T1	$+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (-2 \text{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) - 4 (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + (\lambda_{10})^*))$
T3	$- 2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	$+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (-2 \text{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) - 4 (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + (\lambda_{10})^*))$	$- 2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	$+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (-3 \text{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) - 4 (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + (\lambda_{10})^*))$	$- 3 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 7 of 8

Column	Representative term structure	
T1	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	
T2	$+64\lambda_8\left(\lambda_8\right)^*$	
T3	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+64\lambda_8\left(\lambda_8\right)^*$	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	$+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+64\lambda_8\left(\lambda_8\right)^*$	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	$+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 8 of 8

Column	Representative term structure
T1	$\lambda_{HS_2}^{(A)} (3T_\nu^{(2)} + 6 \operatorname{Tr}\big(Y_d \ (Y_d)^\dagger\big) + 2 \operatorname{Tr}\big(Y_e \ (Y_e)^\dagger\big) + 6 \operatorname{Tr}\big(Y_u \ (Y_u)^\dagger\big))$
T2	$+12\lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	$\lambda_{HS_2}^{(A)} (3T_\nu^{(2)} + 6 \operatorname{Tr}\big(Y_d \ (Y_d)^\dagger\big) + 2 \operatorname{Tr}\big(Y_e \ (Y_e)^\dagger\big) + 6 \operatorname{Tr}\big(Y_u \ (Y_u)^\dagger\big))$	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\lambda_{HS_2}^{(A)} (3T_\nu^{(2)} + 6 \operatorname{Tr}\big(Y_d \ (Y_d)^\dagger\big) + 2 \operatorname{Tr}\big(Y_e \ (Y_e)^\dagger\big) + 6 \operatorname{Tr}\big(Y_u \ (Y_u)^\dagger\big))$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	$\lambda_{HS_2}^{(A)} (3T_\nu^{(2)} + 6 \operatorname{Tr}\big(Y_d \ (Y_d)^\dagger\big) + 2 \operatorname{Tr}\big(Y_e \ (Y_e)^\dagger\big) + 6 \operatorname{Tr}\big(Y_u \ (Y_u)^\dagger\big))$	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	$+12\lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+12\lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	$+12\lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	$+12\lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$

$\beta_{\lambda_7}^{(1)}$

Quantity: λ_7

$$16\pi^2\,\mu\frac{d\lambda_7}{d\mu}=\beta_{\lambda_7}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1	$\lambda_7(2T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	
T2	$+(\lambda_5)^*\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)$	
T3	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_7(2T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$+(\lambda_5)^*\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)$	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_7

Terms 2 of 5

Column	Representative term structure		
T1	$+2\lambda_{11}(\lambda_5)^*$		
T2	$-9g_2^2\lambda_7$		
T3	$-3g_1^2\lambda_7$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_{11}(\lambda_5)^*$	$-9g_2^2\lambda_7$	$-3g_1^2\lambda_7$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_{11}(\lambda_5)^*$	$-9g_2^2\lambda_7$	$-3g_1^2\lambda_7$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_7

Terms 3 of 5

Column		Representative term structure
T1		$+2\lambda_1\lambda_7$
T2		$+8\lambda_{HS_1}^{(1)}\lambda_7$
T3		$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_1\lambda_7$	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_1\lambda_7$	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_7

Terms 4 of 5

Column	Representative term structure	
T1	$+2\lambda_7\lambda_{S_1}^{(1)}$	
T2	$+4\lambda_8\lambda_9$	
T3	$\lambda_7(3T_\nu^{(1)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$	$\lambda_7(3T_\nu^{(1)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_7

Terms 5 of 5

Column	Representative term structure
T1	$+\frac{3}{2}(\lambda_5)^*\text{Tr}\Big(y_2\,(y_1)^\dagger\Big)$

Model	α	S_1	S_2	F	T1
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+\frac{3}{2}(\lambda_5)^*\text{Tr}\Big(y_2\,(y_1)^\dagger\Big)$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—

$\beta_{\lambda_{12}}^{(1)}$

Quantity: λ_{12}

$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 5

Column		Representative term structure	
T1	$\lambda_{HS_1}^{(1)}\left(\frac{1}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11}\right)$		
T2	$+\lambda_{HS_1}^{(A)}\left(-\frac{1}{8}\text{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$		
T3	$+\lambda_{HS_2}^{(1)}\left(\frac{1}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10}\right)$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_{HS_1}^{(1)}\left(\frac{1}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11}\right)$	$+\lambda_{HS_1}^{(A)}\left(-\frac{1}{8}\text{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$	$+\lambda_{HS_2}^{(1)}\left(\frac{1}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10}\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$\lambda_{HS_1}^{(1)}\left(\frac{3}{4}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11}\right)$	$+\lambda_{HS_1}^{(A)}\left(-\frac{3}{16}\text{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$	$+\lambda_{HS_2}^{(1)}\left(\frac{3}{4}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{12}

Terms 2 of 5

Column		Representative term structure
T1	$+\lambda_{HS_2}^{(A)}\left(-\frac{3}{8}\text{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$	
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	
T3	$-4\text{Tr}\left(y_2\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+\lambda_{HS_2}^{(A)}\left(-\frac{3}{8}\text{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-4\text{Tr}\left(y_2\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+\lambda_{HS_2}^{(A)}\left(-\frac{9}{16}\text{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: λ_{12}

Terms 3 of 5

Column		Representative term structure
T1	$+6\lambda_9$	$\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	
T3	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	—	—	—

Quantity: λ_{12}

Terms 4 of 5

Column	Representative term structure
T1	$-3g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$+2\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-3g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-3g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{12}

Terms 5 of 5

Column	Representative term structure
T1	$-6\lambda_7\lambda_5$
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) + 2 \text{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) + 6 \text{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \right)$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-6\lambda_7\lambda_5$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-6\lambda_7\lambda_5$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) + 2 \text{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) + 6 \text{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{\lambda_8}^{(1)}$

Quantity: λ_8

$$16\pi^2\,\mu\frac{d\lambda_8}{d\mu}=\beta_{\lambda_8}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_8(2T_\nu^{(2)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$
T2	$+ (\lambda_5)^*\,\text{Tr}\Big(y_1\,(y_2)^\dagger\Big)$
T3	$+4\lambda_7\,(\lambda_9)^*$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_8(2T_\nu^{(2)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$	$+ (\lambda_5)^*\,\text{Tr}\Big(y_1\,(y_2)^\dagger\Big)$	$+4\lambda_7\,(\lambda_9)^*$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	$+4\lambda_7\,(\lambda_9)^*$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_8

Terms 2 of 4

Column	Representative term structure
T1	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T2	$-9g_2^2\lambda_8$
T3	$-3g_1^2\lambda_8$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	$-9g_2^2\lambda_8$	$-3g_1^2\lambda_8$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	$-9g_2^2\lambda_8$	$-3g_1^2\lambda_8$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_8

Terms 3 of 4

Column		Representative term structure
T1		$+2\lambda_1\lambda_8$
T2		$+8\lambda_{HS_2}^{(1)}\lambda_8$
T3		$+4\lambda_{HS_2}^{(A)}\lambda_8$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_1\lambda_8$	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_1\lambda_8$	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_8

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_8\lambda_{S_2}^{(1)}$
T2	$\lambda_8(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3	$+\frac{3}{2}\left(\lambda_5\right)^*\operatorname{Tr}\left(y_1\left(y_2\right)^\dagger\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_8\lambda_{S_2}^{(1)}$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_8\lambda_{S_2}^{(1)}$	$\lambda_8(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+\frac{3}{2}\left(\lambda_5\right)^*\operatorname{Tr}\left(y_1\left(y_2\right)^\dagger\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

$\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_1}^{(A)}}{d\mu}=\beta_{\lambda_{S_1}^{(A)}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure	
T1	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	
T2	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	
T3	$-\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 2 of 6

Column		Representative term structure
T1		$+ \frac{3g_2^4}{4}$
T2		$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$
T3		$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 3 of 6

Column		Representative term structure
T1		$+\frac{3g_1^4}{2}$
T2		$+\frac{\lambda_{12}^{(1)2}}{4}$
T3		$+\frac{\lambda_{HS_1}^{(1)2}}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	$+\frac{3g_1^4}{2}$	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	$+\frac{3g_1^4}{2}$	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	—	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+\frac{3g_1^4}{2}$	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	—	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+\frac{3g_1^4}{2}$	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 4 of 6

Column		Representative term structure
T1	$+3\lambda_{HS_1}^{(A)^2}$	
T2	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	
T3	$+22\lambda_{S_1}^{(A)^2}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 5 of 6

Column	Representative term structure	
T1	$+g_1^2\left(36g_2^2-6\lambda_{S_1}^{(1)}-48\lambda_{S_1}^{(A)}\right)$	
T2	$+24g_1^4$	
T3	$+\frac{3\lambda_{12}^{(1)2}}{4}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	(1,0)	(3,1)	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(1,-1)	—	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(1,0)	—	—	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(1,1)	—	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	(3,-1)	—	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	(3,0)	—	—	—
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	$(2,-\frac{1}{2})$	—	—	—
T3-D-p0	0	(3,0)	(1,1)	$(2,\frac{1}{2})$	—	—	—
T3-D-p2	2	(3,1)	(1,2)	$(2,\frac{3}{2})$	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	$(2,-\frac{3}{2})+g_1^2\left(36g_2^2-6\lambda_{S_1}^{(1)}-48\lambda_{S_1}^{(A)}\right)$		$+24g_1^4$	$+\frac{3\lambda_{12}^{(1)2}}{4}$
T3-E-m2	-2	(3,-1)	(3,0)	$(2,-\frac{1}{2})$		—	$+\frac{3\lambda_{12}^{(1)2}}{4}$
T3-E-p0	0	(3,0)	(3,1)	$(2,\frac{1}{2})$		—	$+\frac{3\lambda_{12}^{(1)2}}{4}$
T3-E-p2	2	(3,1)	(3,2)	$(2,\frac{3}{2})$		—	$+\frac{3\lambda_{12}^{(1)2}}{4}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 6 of 6

Column	Representative term structure
T1	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T2	$+12\lambda_{12}^{(A)2}$
T3	$-\frac{\lambda_{12}^{(\times)2}}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$

$$\beta_{\lambda_{11}}^{(1)}$$

Quantity: λ_{11}

$$16\pi^2\mu\frac{d\lambda_{11}}{d\mu}=\beta_{\lambda_{11}}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1	$\lambda_{12}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11})$	
T2	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	
T3	$+\lambda_{S_1}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+12\lambda_{11})$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_{12}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	$+\lambda_{S_1}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+12\lambda_{11})$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+\lambda_{12}^{(A)}\left(\frac{1}{8}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{11}

Terms 2 of 5

Column	Representative term structure
T1	$+ \lambda_9 (2 \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + 4 ((\lambda_{10})^* + 5 (\lambda_{11})^*))$
T2	$+ \lambda_{11} (3 T_{\nu}^{(1)} + T_{\nu}^{(2)})$
T3	$- 4 \operatorname{Tr} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \left(y_1 \right)^{\dagger} \right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+ \lambda_9 (2 \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + 4 ((\lambda_{10})^* + 5 (\lambda_{11})^*))$	$+ \lambda_{11} (3 T_{\nu}^{(1)} + T_{\nu}^{(2)})$	$- 4 \operatorname{Tr} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \left(y_1 \right)^{\dagger} \right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+ \lambda_9 (3 \operatorname{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + 4 ((\lambda_{10})^* + 5 (\lambda_{11})^*))$	—	$- 5 \operatorname{Tr} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \left(y_1 \right)^{\dagger} \right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{11}

Terms 3 of 5

Column	Representative term structure	
T1	$-9g_2^2\lambda_{11}$	
T2	$-3g_1^2\lambda_{11}$	
T3	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-9g_2^2\lambda_{11}$	$-3g_1^2\lambda_{11}$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-9g_2^2\lambda_{11}$	$-3g_1^2\lambda_{11}$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{11}

Terms 4 of 5

Column	Representative term structure
T1	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$
T2	$+4\lambda_7\lambda_5$
T3	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_7\lambda_5$	—
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_7\lambda_5$	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	—	—
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	—	—	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	—	—	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	—	—	—

Quantity: λ_{11}

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda_{S_1}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+12\lambda_{11}\right)$
T2	$+\lambda_{11}\left(\frac{9}{2}T_{\nu}^{(1)}+\frac{3}{2}T_{\nu}^{(2)}\right)$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\lambda_{S_1}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+12\lambda_{11}\right)$	$+\lambda_{11}\left(\frac{9}{2}T_{\nu}^{(1)}+\frac{3}{2}T_{\nu}^{(2)}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{\lambda_9}^{(1)}$

Quantity: λ_9

$$16\pi^2\mu\frac{d\lambda_9}{d\mu}=\beta_{\lambda_9}^{(1)}$$

Terms 1 of 6

Column	Representative term structure	
T1	$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	
T2	$+ \lambda_{10}(\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)+2\lambda_{11})$	
T3	$+ \lambda_{11}\ \text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	$+ \lambda_{10}(\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)+2\lambda_{11})$	$+ \lambda_{11}\ \text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_9

Terms 2 of 6

Column	Representative term structure
T1	$-2\operatorname{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$
T2	$+4\lambda_7\left(\lambda_8\right)^*$
T3	$-9g_2^2\lambda_9$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—	—
T3-A-m2	-2	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-A-p0	0	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	—
T3-B-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	—	—	—
T3-B-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	$-2\operatorname{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+4\lambda_7\left(\lambda_8\right)^*$	$-9g_2^2\lambda_9$
T3-B-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	—	—	—
T3-C-m3	-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	—	—	—
T3-C-m1	-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	—	$+4\lambda_7\left(\lambda_8\right)^*$	$-9g_2^2\lambda_9$
T3-C-p1	1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	—	—	—
T3-D-m2	-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-D-p0	0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	—	—	—
T3-D-p2	2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	—	—	—
T3-E-m4	-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	—	—	—
T3-E-m2	-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	—	—	—
T3-E-p0	0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	—	—	—
T3-E-p2	2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	—	—	—

Quantity: λ_9

Terms 3 of 6

Column	Representative term structure
T1	$-3g_1^2\lambda_9$
T2	$+8\lambda_{12}^{(1)}\lambda_9$
T3	$+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$-3g_1^2\lambda_9$	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$-3g_1^2\lambda_9$	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_9

Terms 4 of 6

Column	Representative term structure
T1	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}^2$
T2	$+2\lambda_{S_1}^{(1)}\lambda_9$
T3	$+2\lambda_9\lambda_{S_2}^{(1)}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}^2$	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}^2$	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_9

Terms 5 of 6

Column	Representative term structure	
T1	$+5\lambda_{10}^2$	
T2	$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	
T3	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+2\lambda_{11}\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+5\lambda_{10}^2$ $+5\lambda_{11}^2$	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+5\lambda_{10}^2$ $+5\lambda_{11}^2$	$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+2\lambda_{11}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_9

Terms 6 of 6

Column	Representative term structure
T1	$+\frac{3}{2}\lambda_{11}\text{Tr}\Big(y_2\left(y_1\right)^\dagger\Big)$
T2	$-\frac{5}{2}\text{Tr}\Big(y_2\left(y_1\right)^\dagger y_2\left(y_1\right)^\dagger\Big)$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\frac{3}{2}\lambda_{11}\text{Tr}\Big(y_2\left(y_1\right)^\dagger\Big)$	$-\frac{5}{2}\text{Tr}\Big(y_2\left(y_1\right)^\dagger y_2\left(y_1\right)^\dagger\Big)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{\lambda_{10}}^{(1)}$

Quantity: λ_{10}

$16\pi^2\mu\frac{d\lambda_{10}}{d\mu}=\beta_{\lambda_{10}}^{(1)}$

Terms 1 of 5

Column		Representative term structure
T1	$\lambda_{12}^{(1)}\langle\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11}\rangle$	
T2	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$	
T3	$+\lambda_9(2\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\ (\lambda_{10})^*+(\lambda_{11})^*))$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$\lambda_{12}^{(1)}\langle\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11}\rangle$	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$	$+\lambda_9(2\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\ (\lambda_{10})^*+(\lambda_{11})^*))$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	$+\lambda_{12}^{(A)}\left(\frac{1}{8}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$	$+\lambda_9(3\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\ (\lambda_{10})^*+(\lambda_{11})^*))$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{10}

Terms 2 of 5

Column	Representative term structure	
T1	$+\lambda_{S_2}^{(1)}(\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+12\lambda_{10})$	
T2	$+\lambda_{10}(T_\nu^{(1)}+3T_\nu^{(2)})$	
T3	$-4\text{Tr}\left(y_2\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+\lambda_{S_2}^{(1)}(\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+12\lambda_{10})$	$+\lambda_{10}(T_\nu^{(1)}+3T_\nu^{(2)})$	$-4\text{Tr}\left(y_2\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	$-5\text{Tr}\left(y_2\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{10}

Terms 3 of 5

Column	Representative term structure		
T1	$+4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$		
T2	$-9g_2^2\lambda_{10}$		
T3	$-3g_1^2\lambda_{10}$		

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{10}$	$-3g_1^2\lambda_{10}$
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{10}$	$-3g_1^2\lambda_{10}$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{10}

Terms 4 of 5

Column	Representative term structure
T1	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	—

Quantity: λ_{10}

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda_{S_2}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+12\lambda_{10}\right)$
T2	$+\lambda_{10}\left(\frac{3}{2}T_\nu^{(1)}+\frac{9}{2}T_\nu^{(2)}\right)$

Model	α	S_1	S_2	F	T1	T2
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	$+\lambda_{S_2}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+12\lambda_{10}\right)$	$+\lambda_{10}\left(\frac{3}{2}T_\nu^{(1)}+\frac{9}{2}T_\nu^{(2)}\right)$
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—

$\beta_{\lambda_{S_2}^{(A)}}^{(1)}$

Quantity: $\lambda_{S_2}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_2}^{(A)}}{d\mu}=\beta_{\lambda_{S_2}^{(A)}}^{(1)}$$

Terms 1 of 6

Column		Representative term structure
T1	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	
T2	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	
T3	$-\text{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 2 of 6

Column		Representative term structure
T1	$+ \frac{3g_2^4}{4}$	
T2	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_2}^{(1)}}{2} - 12\lambda_{S_2}^{(A)} \right)$	
T3	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_2}^{(1)}}{2} - 12\lambda_{S_2}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$
T3-A-m2	-2	(1, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$
T3-A-p0	0	(1, 0)	(3, 1)	$(2, \frac{1}{2})$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_2}^{(1)}}{2} - 12\lambda_{S_2}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$
T3-B-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(1, -1)	—	—	—
T3-B-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(1, 0)	—	—	—
T3-B-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(1, 1)	—	—	—
T3-C-m3	-3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	(3, -1)	—	—	—
T3-C-m1	-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	(3, 0)	—	—	—
T3-C-p1	1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	$(2, -\frac{1}{2})$	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	$(2, \frac{1}{2})$	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	$(2, \frac{3}{2})$	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	$(2, -\frac{3}{2})$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_2}^{(1)}}{2} - 12\lambda_{S_2}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$
T3-E-m2	-2	(3, -1)	(3, 0)	$(2, -\frac{1}{2})$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$
T3-E-p0	0	(3, 0)	(3, 1)	$(2, \frac{1}{2})$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_2}^{(1)}}{2} - 12\lambda_{S_2}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$
T3-E-p2	2	(3, 1)	(3, 2)	$(2, \frac{3}{2})$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 3 of 6

Column		Representative term structure
T1		$+\frac{3g_1^4}{2}$
T2		$+\frac{\lambda_{12}^{(1)2}}{4}$
T3		$+\frac{\lambda_{HS_2}^{(1)2}}{2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3g_1^4}{2}$	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3g_1^4}{2}$	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+\frac{3g_1^4}{2}$	—	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+\frac{3g_1^4}{2}$	—	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	—	—	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 4 of 6

Column	Representative term structure
T1	$+3\lambda_{HS_2}^{(A)^2}$
T2	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$
T3	$+22\lambda_{S_2}^{(A)^2}$

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+22\lambda_{S_2}^{(A)^2}$
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+22\lambda_{S_2}^{(A)^2}$
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+22\lambda_{S_2}^{(A)^2}$
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+22\lambda_{S_2}^{(A)^2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+22\lambda_{S_2}^{(A)^2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+22\lambda_{S_2}^{(A)^2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+3\lambda_{HS_2}^{(A)^2}$	$+\frac{7\lambda_{S_2}^{(1)^2}}{4}$	$+22\lambda_{S_2}^{(A)^2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 5 of 6

Column	Representative term structure			
T1	$+ \frac{3\lambda_{12}^{(1)2}}{4}$			
T2	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$			
T3	$+ 12 \lambda_{12}^{(A)2}$			

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1, -2)	(3, -1)	(2, - $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1, -1)	(3, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(1, -1)	—	—	—
T3-B-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	—	—	—
T3-C-m3	-3	(2, - $\frac{3}{2}$)	(2, - $\frac{1}{2}$)	(3, -1)	—	—	—
T3-C-m1	-1	(2, - $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	—	—	—
T3-D-m2	-2	(3, -1)	(1, 0)	(2, - $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3, -2)	(3, -1)	(2, - $\frac{3}{2}$)	$+ \frac{3\lambda_{12}^{(1)2}}{4}$	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	$+ 12 \lambda_{12}^{(A)2}$
T3-E-m2	-2	(3, -1)	(3, 0)	(2, - $\frac{1}{2}$)	$+ \frac{3\lambda_{12}^{(1)2}}{4}$	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	$+ 12 \lambda_{12}^{(A)2}$
T3-E-p0	0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	$+ \frac{3\lambda_{12}^{(1)2}}{4}$	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	$+ 12 \lambda_{12}^{(A)2}$
T3-E-p2	2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	$+ \frac{3\lambda_{12}^{(1)2}}{4}$	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	$+ 12 \lambda_{12}^{(A)2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 6 of 6

Column	Representative term structure	
T1	$-\frac{\lambda_{12}^{(\times)^2}}{2}$	
T2	$+g_1^2\left(36g_2^2-6\lambda_{S_2}^{(1)}-48\lambda_{S_2}^{(A)}\right)$	
T3	$+24g_1^4$	

Model	α	S_1	S_2	F	T1	T2	T3
T3-A-m4	-4	(1,-2)	(3,-1)	(2,- $\frac{3}{2}$)	—	—	—
T3-A-m2	-2	(1,-1)	(3,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-A-p0	0	(1,0)	(3,1)	(2, $\frac{1}{2}$)	—	—	—
T3-B-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(1,-1)	—	—	—
T3-B-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1,0)	—	—	—
T3-B-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1,1)	—	—	—
T3-C-m3	-3	(2,- $\frac{3}{2}$)	(2,- $\frac{1}{2}$)	(3,-1)	—	—	—
T3-C-m1	-1	(2,- $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3,0)	—	—	—
T3-C-p1	1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3,1)	—	—	—
T3-D-m2	-2	(3,-1)	(1,0)	(2,- $\frac{1}{2}$)	—	—	—
T3-D-p0	0	(3,0)	(1,1)	(2, $\frac{1}{2}$)	—	—	—
T3-D-p2	2	(3,1)	(1,2)	(2, $\frac{3}{2}$)	—	—	—
T3-E-m4	-4	(3,-2)	(3,-1)	(2,- $\frac{3}{2}$)	$-\frac{\lambda_{12}^{(\times)^2}}{2}$	—	—
T3-E-m2	-2	(3,-1)	(3,0)	(2,- $\frac{1}{2}$)	$-\frac{\lambda_{12}^{(\times)^2}}{2}$	—	—
T3-E-p0	0	(3,0)	(3,1)	(2, $\frac{1}{2}$)	$-\frac{\lambda_{12}^{(\times)^2}}{2}$	—	—
T3-E-p2	2	(3,1)	(3,2)	(2, $\frac{3}{2}$)	$-\frac{\lambda_{12}^{(\times)^2}}{2}$	$+g_1^2\left(36g_2^2-6\lambda_{S_2}^{(1)}-48\lambda_{S_2}^{(A)}\right)$	$+24g_1^4$